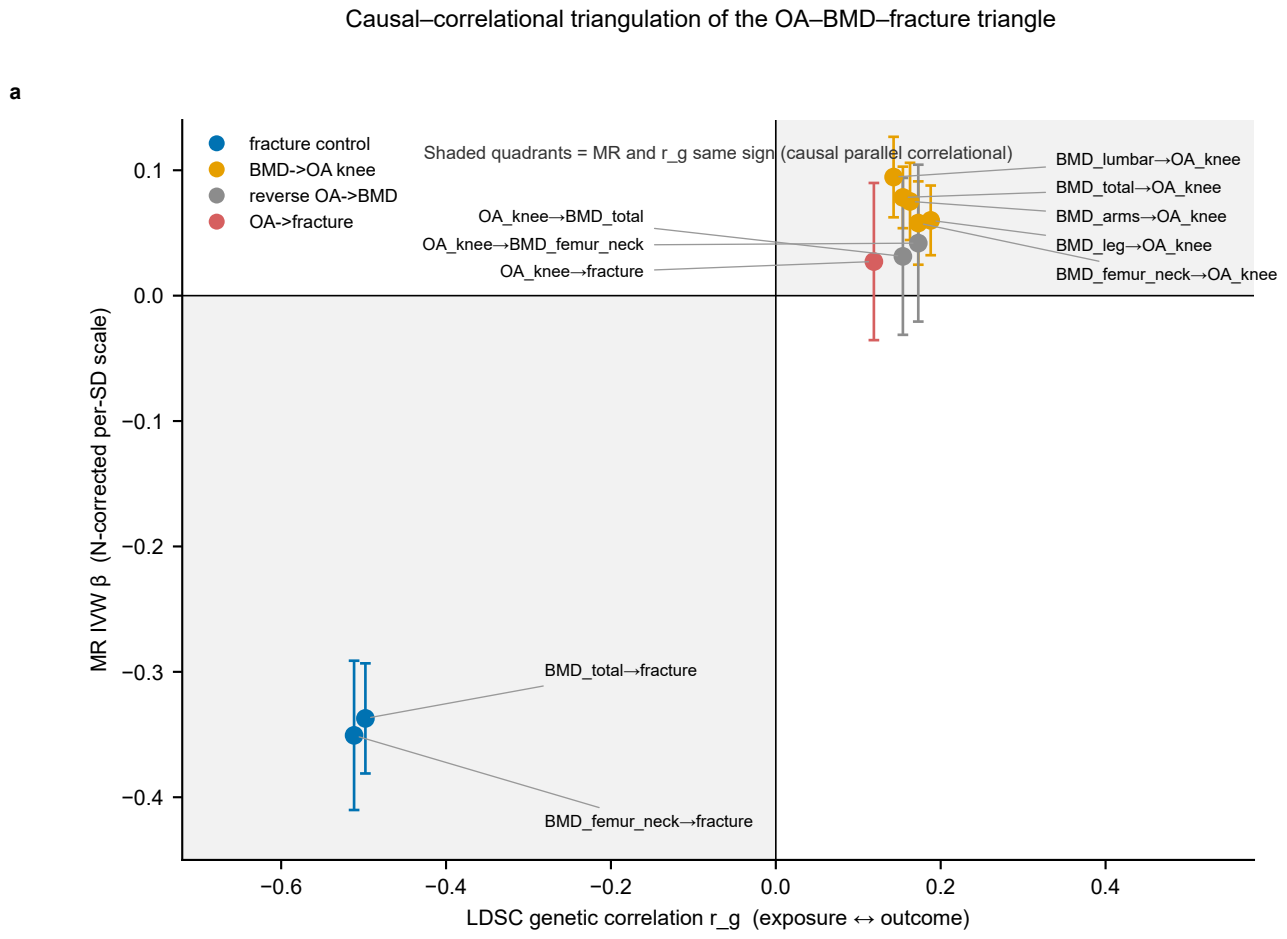
Causal–correlational triangulation of the OA–BMD–fracture triangle
a
fracture control
BMD->OA knee
reverse OA->BMD
OA->fracture
Shaded quadrants = MR and r_g same sign (causal parallel correlational)
BMD_lumbar→OA_knee
BMD_total→OA_knee
BMD_arms→OA_knee
BMD_leg→OA_knee
BMD_femur_neck→OA_knee
OA_knee→BMD_total
OA_knee→BMD_femur_neck
OA_knee→fracture
BMD_total→fracture
BMD_femur_neck→fracture
MR IVW β (N-corrected per-SD scale)
LDSC genetic correlation r_g (exposure ↔ outcome)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.6
−0.4
−0.2
0.0
0.2
0.4